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Anticompetitive Effects of Horizontal Acquisitions: The Impact of Within-Industry Product Similarity

Contents

1 Big picture

- **Research question.** Do horizontal acquisitions reduce competition and raise market power when the industry has both high concentration and high product similarity?
- **Core idea.** Prior empirical work often asks whether horizontal mergers are anticompetitive using only concentration measures. This paper argues that concentration is informative only when products are sufficiently similar; in differentiated-product industries, competition is localized and concentration ratios are weak proxies for market power.
- **New measure.** The paper constructs **Industry Product Similarity** (IPS), a three-digit SIC industry-year measure based on textual similarity in firms' 10-K business descriptions. IPS ranges from zero to one. Higher IPS means that firms in the same industry offer products with more similar features.
- **Main prediction.** Horizontal acquisitions should be especially anticompetitive when **IPS is high** and **industry concentration is high**. The intuition is that product homogeneity makes coordination easier, while concentration reduces the number of firms that must coordinate.
- **Data setting.** The empirical tests use U.S. public acquirers and U.S. targets from SDC over 1996–2015, CRSP/Compustat firm data, BEA input-output links for customers and suppliers, Census concentration measures, BLS producer price indices, and hand-collected FTC/DOJ challenge data.
- **Validation result.** IPS is positively correlated with other product-market similarity or homogeneity measures and negatively correlated with price-cost margins on average, which is consistent with product similarity intensifying price competition absent coordination.
- **Acquisition likelihood result.** Higher IPS raises the probability that a firm makes a horizontal acquisition, especially in concentrated industries. IPS has no comparable effect on cross-industry acquisitions.
- **Announcement-return result.** Acquirer and rival announcement returns are higher when horizontal acquisitions occur in high-IPS industries, and the effect is strongest when the industry is concentrated.
- **Target premium and combined-gain result.** Targets receive higher offer premiums and acquirer-target combined wealth gains are higher in high-IPS/high-concentration horizontal acquisitions.
- **Product-market result.** Industry prices rise after horizontal acquisitions when IPS interacts with high concentration, customer dependency, or target size.
- **Customer and supplier result.** Dependent customer and supplier firms have negative announcement returns when the horizontal acquisition occurs in high-IPS/high-concentration industries, consistent with expected harm through market power.
- **Regulatory result.** The FTC/DOJ is more likely to challenge horizontal acquisitions in high-IPS industries, particularly when the deal occurs in a concentrated industry or when customer/supplier dependency is high.
- **Heterogeneity result.** The evidence is weaker in IP-focused industries and stronger in commodity-based industries, where product similarity is naturally more relevant for coordination.
- **Bottom line.** The paper reconciles mixed prior evidence on horizontal mergers by showing that concentration alone is incomplete. Product similarity is the missing conditioning variable.

2 Conceptual framework and empirical predictions

2.1 Why concentration alone is not enough

The traditional antitrust logic says that a horizontal merger reduces the number of competitors and therefore should reduce competition. However, concentration alone is a blunt measure. In an industry with highly differentiated products, two firms may be in the same three-digit SIC code but sell products that are not close substitutes. In that case, a merger may not meaningfully alter the competitive pressure faced by most firms in the industry.

- Concentration measures such as the four-firm ratio and Herfindahl index summarize market shares.
- They do not directly measure whether products are close substitutes.
- Antitrust guidelines emphasize that product homogeneity and standardization can facilitate coordination, while product heterogeneity can impede it.
- The paper's key argument is that the likelihood of an anticompetitive horizontal acquisition depends on the **joint** level of concentration and product similarity.

Interpretation: If products are differentiated, competition can be local to particular product niches, and industry-wide concentration may exaggerate the degree of market power. If products are homogeneous, competitors are closer substitutes and tacit coordination becomes easier.

2.2 Industry product similarity, interdependence, and coordination

In standard Cournot and Bertrand models with homogeneous products, firms' pricing and output choices are tightly linked. With few firms, similar products, and observable behavior, firms can learn that aggressive competition triggers rival responses. The paper connects this logic to Chamberlin-style interdependence and to the broader industrial organization literature on tacit coordination.

- High product similarity reduces the number of dimensions on which firms must coordinate.
- High concentration reduces the number of firms involved in coordination.
- Inelastic demand increases the payoff from coordinated price increases.
- Horizontal acquisitions can make coordination easier by removing a competitor, changing market shares, and increasing the returns to market power.

Interpretation: The paper does not argue that high similarity always reduces competition. Product similarity can intensify price competition under Bertrand logic. The point is more subtle: high similarity combined with concentration and a horizontal acquisition can move the industry toward coordination and market power.

2.3 Horizontal acquisitions as a tool to reduce competition

A horizontal acquisition combines firms in the same industry. Under the competition-reduction hypothesis, the acquisition changes the competitive environment in a way that benefits the acquirer, the target, and rivals, but harms customers and suppliers.

- The acquirer can gain from higher future prices or reduced rivalry.
- The target can demand a premium because some of the market-power gain is paid to target shareholders.
- Rivals can gain because they also benefit from a less competitive industry.

- Dependent customers can lose because input prices rise.
- Dependent suppliers can lose because more powerful downstream firms can obtain better terms.
- Antitrust authorities should be more likely to challenge such deals.

Interpretation: A key empirical signature of reduced competition is that rivals gain. If a merger only creates firm-specific synergies, rivals should not benefit. Positive rival returns point toward industry-wide market-power effects.

2.4 Main empirical predictions

The paper's conceptual discussion leads to four central predictions.

1. IPS is positively related to the likelihood of a horizontal acquisition, especially in concentrated industries.
2. Acquirer returns, rival returns, target premiums, combined wealth gains, post-merger price increases, and challenge probabilities are increasing in IPS, especially in concentrated industries.
3. Dependent customer and supplier returns are decreasing in IPS for horizontal acquisitions, especially when concentration is high.
4. The effect of IPS is stronger when the target is larger or when dependent customer demand is less elastic.

Interpretation: These predictions distinguish the competition-reduction hypothesis from alternative explanations. Synergies may help acquirers and targets, but they do not naturally predict higher rival returns, negative dependent-customer and supplier returns, price increases, or more antitrust challenges.

3 Data, sample construction, and measurement

3.1 Firm-level data for IPS validation

The validation tests use CRSP daily stock returns and CRSP/Compustat accounting data for fiscal years 1996–2015. The sample excludes firms that are not traded on NYSE, NASDAQ, or AMEX, firms missing sales or total assets, financial firms, utility firms, and miscellaneous industry classifications.

- CRSP provides returns used to construct industry homogeneity and event-study abnormal returns.
- Compustat provides sales, assets, operating income, leverage, R&D, advertising, capital expenditures, and related controls.
- Industries are defined at the three-digit SIC level.

Interpretation: The sample period is chosen because it overlaps with 10-K text availability, modern SDC transaction coverage, and data for validation and outcome measures.

3.2 Mergers and acquisitions sample

The merger sample comes from SDC Mergers and Acquisitions data.

- Acquirers are publicly traded U.S. firms.
- Targets are U.S. public firms, private firms, or subsidiaries.
- Deals must be completed and have value of at least $1 million.
- The acquirer must own less than 50% of the target before the deal and obtain at least 50% of target shares in the transaction.
- A horizontal acquisition is one in which acquirer and target share the same three-digit primary SIC code.
- A cross-industry acquisition is one in which they do not share the same three-digit primary SIC code.

Interpretation: Including private and subsidiary targets matters because public-target deals are only a small part of the acquisition market. Several tests use public targets only because target returns and offer premiums are observable for those deals.

3.3 Industry concentration

The paper uses U.S. Census concentration data.

- The four-firm ratio is the combined market share of the four largest firms in an industry.
- The Herfindahl index is available for manufacturing industries.
- The paper uses the four-firm ratio in main tests because it is available for nearly all industries.
- An industry is classified as highly concentrated if the four largest firms account for at least 60% of total industry sales.

Interpretation: This concentration measure is close to antitrust practice because it uses Census data that include both public and private firms. Compustat-only concentration measures would miss private firms and can be noisy.

3.4 Construction of Industry Product Similarity

The paper constructs IPS from textual analysis of firms' 10-K business descriptions. Let $\boldsymbol{x}_{j,t}$ be the vector of product words used by firm j in year t. For industry k, let $\boldsymbol{m}_{k,t}$ be the vector measuring how frequently words appear across firms in the industry. The firm-industry product similarity can be written as a cosine similarity:

$$Sim_{j,k,t} = \frac{\boldsymbol{x}'_{j,t}\boldsymbol{m}_{k,t}}{\|\boldsymbol{x}_{j,t}\|\|\boldsymbol{m}_{k,t}\|}. \tag{1}$$

The industry-year measure is the average across firms in the industry:

$$IPS_{k,t} = \frac{1}{N_{k,t}} \sum_{j \in k} Sim_{j,k,t}. \tag{2}$$

- IPS lies between zero and one.
- Higher IPS means the products offered by firms in the industry are more alike.
- IPS is calculated for three-digit SIC industries, not text-based industries, because the goal is to measure similarity within conventional industrial classifications.
- Miscellaneous industries and industries with too few firms are excluded.

Interpretation: IPS combines SIC production-process similarity with textual product-feature similarity. It is designed to identify industries where firms are classified similarly and also describe their products similarly.

3.5 Customer and supplier dependency

The paper uses BEA benchmark input-output tables to measure dependence between industries.

- **Customer Input Coefficient (CIC):** for a customer-acquirer industry pair, the dollar amount of the acquirer industry's output purchased by the customer industry divided by the customer industry's total output.
- **Supplier Percentage Sold (SPS):** for a supplier-acquirer industry pair, the share of the supplier industry's output sold to the acquirer industry.
- For each acquirer industry, the paper identifies the top five dependent customer industries with the highest CIC and the top five dependent supplier industries with the highest SPS.

Interpretation: These measures capture demand inelasticity and bargaining exposure. If customers depend heavily on the acquirer industry, a price increase is more harmful. If suppliers depend heavily on the acquirer industry, stronger downstream market power can hurt suppliers.

3.6 Outcome variables

- **Acquisition likelihood:** firm-year indicator for making a horizontal acquisition; deal-level indicator for whether an acquisition is horizontal.
- **Acquirer CAR:** three-day cumulative abnormal return around the acquisition announcement.
- **Rival CAR:** average announcement return of non-acquiring firms in the same three-digit SIC industry.
- **Target offer premium:** log initial offer price relative to target share price before the announcement.
- **Combined wealth gain:** value-weighted or equal-weighted gain of acquirer and target.

- **Industry price changes:** two-year post-acquisition changes in PPI inflation for the acquirer's three-digit SIC industry.
- **Dependent customer/supplier returns:** event-window returns of firms in industries that rely on or sell to the acquirer industry.
- **Performance changes:** changes in price-cost margins and expense ratios for acquirers relative to matched controls.
- **Regulatory challenge:** indicator for FTC/DOJ challenge to a proposed acquisition.

Interpretation: The paper triangulates market power through many outcomes. The strongest evidence is not any single regression, but the joint pattern: acquirers, targets, and rivals gain; customers and suppliers lose; prices rise; and regulators challenge more deals.

4 Empirical models and specifications

4.1 Acquisition-likelihood model

A simplified version of the firm-year acquisition model is:

$$HAcq_{i,t} = \alpha_i + \lambda_t + \beta_1 IPS_{k,t} + \beta_2 HighConc_{k,t} + \beta_3(IPS_{k,t} \times HighConc_{k,t}) + \Gamma' X_{i,t} + \varepsilon_{i,t}, \tag{3}$$

where $HAcq_{i,t}$ is an indicator for firm i making a horizontal acquisition, k is the firm's industry, and $X_{i,t}$ includes liquidity, profitability, valuation, R&D, asset turnover, size, and other controls.

Interpretation: β_1 asks whether firms in high-similarity industries are more likely to make horizontal acquisitions. β_3 asks whether this effect is stronger when the industry is concentrated.

4.2 Announcement-return model

For acquirers, rivals, customers, or suppliers, the generic event-study regression is:

$$CAR_d = \alpha + \beta_1 IPS_{k,t} + \beta_2 HighConc_{k,t} + \beta_3(IPS_{k,t} \times HighConc_{k,t}) + \Gamma' X_d + u_d, \tag{4}$$

where d indexes acquisition deals.

Interpretation: Under the competition-reduction hypothesis, β_3 should be positive for acquirer and rival returns but negative for dependent customers and suppliers.

4.3 Target premium and combined wealth gain model

For public-target horizontal acquisitions, the paper estimates:

$$Premium_d = \alpha + \beta_1 IPS_{k,t} + \beta_2 HighConc_{k,t} + \beta_3(IPS_{k,t} \times HighConc_{k,t}) + \Gamma' X_d + u_d, \tag{5}$$

with analogous regressions for value-weighted and equal-weighted combined wealth gains.

Interpretation: If the acquisition generates market power, the target can bargain for a share of the rents. A higher premium in high-IPS/high-concentration industries supports that interpretation.

4.4 Industry price model

The industry-price outcome is the two-year change in producer prices for the acquirer industry:

$$\Delta Price_{k,d} = \alpha + \beta_1 IPS_{k,t} + \beta_2 HighConc_{k,t} + \beta_3(IPS_{k,t} \times HighConc_{k,t}) + \Gamma' X_d + u_d. \tag{6}$$

Interpretation: Price changes are especially important because they move from stock-market expectations to real product-market outcomes. A positive interaction of IPS and concentration directly supports the claim that some mergers raise prices.

4.5 Dependent customer and supplier model

The dependent customer/supplier CAR tests use:

$$CAR_d^{dep} = \alpha + \beta_1 IPS_{k,t} + \beta_2 HighConc_{k,t} + \beta_3(IPS_{k,t} \times HighConc_{k,t}) + \beta_4(IPS_{k,t} \times Exposure_k) + \Gamma' X_d + u_d. \tag{7}$$

Interpretation: The most compelling market-power prediction is negative returns for exposed trading partners. Synergy stories do not naturally predict harm to customers and suppliers.

4.6 Acquirer performance model

To separate market power from operational synergies, the paper matches each acquirer to a non-acquiring firm in the same industry and estimates changes in performance variables:

$$\Delta Y_{j,t:t+2} = \alpha + \beta(IPS_{k,t} \times Acquirer_j) + \delta IPS_{k,t} + \theta Acquirer_j + \Gamma' X_j + u_{j,t}. \tag{8}$$

The dependent variables include price-cost margin, cost of goods sold/sales, SG&A/sales, capital expenditures/assets, R&D/sales, and advertising/sales.

Interpretation: If gains are mostly due to synergies, expense ratios should decline strongly. If gains are due to market power, margins should rise even without broad cost reductions.

4.7 FTC/DOJ challenge model

For public-target horizontal acquisitions, the challenge model is:

$$Challenge_d = \alpha + \beta_1 IPS_{k,t} + \beta_2 HighConc_{k,t} + \beta_3(IPS_{k,t} \times HighConc_{k,t}) + \Gamma' X_d + u_d. \tag{9}$$

Interpretation: The challenge tests ask whether regulators recognize the same anticompetitive features that financial markets price. The paper also interacts IPS with adjusted acquirer returns, rival returns, and combined wealth gains to ask whether high market-power rents make challenges more likely.

5 Identification and credibility

5.1 What the empirical design can identify

The paper is not a randomized experiment. Its empirical contribution is to condition horizontal-merger effects on a theoretically motivated and newly measured industry characteristic: product similarity. The estimates document whether outcomes associated with market power are systematically stronger when IPS and concentration are high.

Interpretation: The empirical design supports a structured interpretation rather than a clean causal design. The authors are explicit that they do not have exogenous variation in competition intensity.

5.2 Why the evidence is stronger than one regression

The credibility comes from the consistency of many separate outcomes.

- Firms in high-IPS industries are more likely to make horizontal, not cross-industry, acquisitions.
- Acquirers and rivals gain when deals occur in high-IPS/high-concentration industries.
- Targets receive higher premiums.
- Industry prices rise.
- Dependent customers and suppliers lose.
- Antitrust authorities challenge more deals.
- Effects are stronger in commodity industries and weaker in IP-focused industries.

Interpretation: This pattern is hard to reconcile with pure synergy, pure diversification, or generic acquisition activity. It matches the competition-reduction hypothesis.

5.3 Key controls and fixed effects

The regressions use controls for acquirer size, target relative size, payment method, target public/private/subsidiary status, industry sales, liquidity, profitability, leverage, valuation, R&D, asset turnover, year fixed effects, firm fixed effects, and/or industry fixed effects depending on the setting.

Interpretation: The controls reduce concerns that IPS simply proxies for firm size, liquidity, valuation, or industry scale. However, unobserved time-varying industry conditions can still matter.

5.4 Main limitations

- There is no exogenous shock to competition intensity.
- There is no exogenous shock to customer or supplier bargaining power.
- SIC industries are imperfect market definitions.
- IPS is based on textual descriptions and may capture both product similarity and disclosure style.
- Event-study outcomes reflect investor expectations, not direct proof of collusion.
- Industry price data are measured at the industry level and may not isolate every product market affected by a merger.

Interpretation: The paper's claim is best read as strong evidence that product similarity is a key conditioning variable for anticompetitive horizontal acquisition effects, not as a randomized proof that every high-IPS merger reduces competition.

6 Tables

6.1 Table 1: IPS validation

Read:

- IPS is negatively correlated with industry price-cost margins: Pearson correlation −0.3690 and Spearman correlation −0.3868.
- IPS is positively correlated with industry homogeneity: Pearson 0.2251 and Spearman 0.2421.
- IPS is positively correlated with the four-firm ratio and Herfindahl index, but these correlations are not so high that IPS is redundant with concentration.
- IPS is also positively correlated with Hoberg-Phillips product market fluidity and firm-pair similarity.

Economic message: On average, similar products intensify price competition, which explains the negative margin correlation. But in concentrated industries, that same similarity can facilitate coordination after a horizontal acquisition.

Table 1
Correlation coefficients of IPS with other industry product market measures.

This table reports the correlation coefficients of IPS with various industry product market measures over the 1996–2015 sample period. Pearson correlation coefficients are shown in the lower triangle and Spearman rank correlations appear above the diagonal. Industry is defined at the three-digit SIC level. Industry price-cost margin is the average value of sales minus cost of goods sold scaled by sales across all firms within the same industry-year. Four-firm ratio and Herfindahl Index are from the US Census website. The Herfindahl Index is only available for the manufacturing sector and therefore the correlation coefficients reported for it are only for firms within this sector. Following Parrino (1997), industry homogeneity is obtained by first regressing firms' daily stock returns in a three-digit SIC industry during a fiscal year on the CRSP daily equally weighted return and the industry daily return, where industry daily return is calculated as the average daily return of all the firms within the same three-digit SIC code as the firm. Industry homogeneity is the average of the firm-level regression coefficients on industry return across all the firms with the same three-digit SIC code. Product market fluidity and firm-pair similarity scores are from the Hoberg-Phillips Data Library. Industry firm-pair similarity is the average similarity score across all pairs that reside within the same industry-year and industry product market fluidity is the average product market fluidity of firms within the same industry-year. All the reported correlation coefficients are significant at the 1% level.

Pearson/Spearman correlation matrix							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) IPS		−0.3868	0.2421	0.2263	0.2285	0.2120	0.4266
(2) Industry price-cost margin	−0.3690		−0.0182	−0.0698	−0.3186	−0.0935	−0.3651
(3) Industry homogeneity	0.2251	−0.2149		0.2081	0.0877	0.3063	0.1201
(4) Four-firm ratio	0.1787	−0.1311	0.1901		0.8075	0.1605	0.1712
(5) Herfindahl Index	0.1548	−0.2252	0.0716	0.8071		0.1071	0.3847
(6) Industry-level product market fluidity	0.2979	−0.4645	0.2954	0.1021	0.0794		0.2512
(7) Industry-firm-pair similarity score	0.3196	−0.2588	−0.0158	0.1054	0.2776	0.2699	

supplier industries using the benchmark input output ta- tration when fewer firms are able to survive (see, e.g.

Table 2
The likelihood of making a horizontal acquisition and IPS.

This table examines the effect of IPS on the likelihood that a firm makes a horizontal or cross-industry acquisition over the 1996–2015 sample period. Mergers and acquisitions data are from SDC. Acquirers are publicly traded firms in the US and targets are publicly traded or privately held firms in the US or subsidiaries of US firms. The deal value is required to be at least $1 million to be included in the sample. All deals are completed, acquirers' percentage of shares owned before the transaction is less than 50%, and acquirers obtain at least 50% of the shares of the target in the transaction. The sample of horizontal (cross-industry) acquisitions includes acquisitions where the acquirer and the target have (do not have) the same three-digit primary SIC code. The dependent variable in Models (1) and (2) is an indicator equal to one if a firm-year observation has at least one horizontal acquisition, and zero otherwise. In Models (3) and (4) the dependent variable is an indicator equal to one if an acquisition is a horizontal acquisition, and zero if it is a cross-industry acquisition. In Model (5) the dependent variable is an indicator equal to one if a firm-year observation has at least one cross-industry acquisition, and zero otherwise. Following Eckbo (1985), high industry concentration is defined by whether the US Census reports that the four firms with the largest sales market shares in an industry account for at least 60% of the total sales of the industry. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

	Horizontal acquisition				Cross-industry acquisition
	Unconditional on making an acquisition		Conditional on making an acquisition		Unconditional on making an acquisition
	(1)	(2)	(3)	(4)	(5)
IPS	0.3091***	0.3205**	0.8530**	0.8253*	0.0837
	(0.006)	(0.010)	(0.028)	(0.063)	(0.331)
High industry concentration		−0.1215*		−0.6199	
		(0.088)		(0.230)	
IPS × High industry concentration		0.2644*		1.2177	
		(0.060)		(0.210)	
Cash/assets	0.0775***	0.0852***	0.0512	0.0649	0.0741***
	(0.002)	(0.003)	(0.224)	(0.145)	(0.000)
NWC/assets	0.0496***	0.0452**	−0.0638	−0.0829	0.0217
	(0.003)	(0.010)	(0.562)	(0.483)	(0.127)
Operating income/assets	0.0717***	0.0713***	0.2403***	0.2441***	0.0401***
	(0.000)	(0.000)	(0.003)	(0.008)	(0.009)
Debt/equity	−0.0155***	−0.0164***	0.0312	0.0292	−0.0135***
	(0.000)	(0.000)	(0.171)	(0.329)	(0.000)
Market-to-book assets	0.0064***	0.0061***	0.0031	0.0026	0.0042***
	(0.000)	(0.000)	(0.264)	(0.302)	(0.001)
R&D/sales	−0.0028**	−0.0023**	0.1460***	0.1355***	−0.0025***
	(0.024)	(0.026)	(0.000)	(0.000)	(0.000)
Asset turnover	−0.0170***	−0.0158**	−0.0264	−0.0292*	−0.0033
	(0.005)	(0.017)	(0.101)	(0.079)	(0.568)
Log (book assets)	−0.0174***	−0.0134**	−0.0113**	−0.0126**	−0.0086*
	(0.005)	(0.016)	(0.028)	(0.030)	(0.055)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	No	No	Yes
Industry fixed effects	No	No	Yes	Yes	No
Observations	52,309	43,982	9,803	8,201	52,309
Adjusted R-squared	0.1521	0.1459	0.1763	0.1545	0.1131

6.2 Table 2: IPS and the likelihood of horizontal acquisitions

Read:

- IPS positively affects the probability of making a horizontal acquisition: coefficient 0.3091 in Model 1.
- Adding high concentration and the interaction preserves the positive IPS effect and produces a positive interaction coefficient 0.2644.
- Conditional on making an acquisition, IPS raises the likelihood that the acquisition is horizontal: coefficient 0.8530 in Model 3.
- IPS does not significantly affect the likelihood of making a cross-industry acquisition.
- The paper reports that a one-standard-deviation increase in IPS raises the unconditional likelihood of a horizontal acquisition by 16.8%, and by 25.2% in concentrated industries.

Economic message: Firms in industries with similar products are more likely to consolidate horizontally. The absence of an effect on cross-industry acquisitions rules out a generic acquisition-propensity explanation.

6.3 Table 3: Acquirer and rival announcement returns

Read:

- Panel A shows that IPS raises acquirer CARs in horizontal acquisitions: coefficient 0.2651 in Model 1.
- The high-concentration interaction is large and positive: 1.5745 in Panel A, Model 2.
- IPS has no comparable effect in cross-industry acquisitions.
- Panel B shows that IPS also raises rival CARs: coefficient 0.0292 in Model 1.
- The rival return effect is driven by concentrated industries: $IPS \times HighConc = 0.1080$.

Economic message: Positive acquirer and rival returns are consistent with expected market-power gains. Rival gains are especially important because they are hard to explain with acquirer-only synergies.

Table 3
Acquirer and rival firm acquisition announcement returns and IPS.

This table examines the effect of IPS on acquirers' and rival firms' acquisition announcement returns when acquirers make a horizontal acquisition, as well as when they make a cross-industry acquisition, over the 1996–2015 sample period. The sample in Panel A, Models (1) to (5) and in Panel B includes horizontal acquisitions where the acquirer and the target share the same three-digit primary SIC code. The sample in Panel A, Model (6), includes cross-industry acquisitions where the acquirer and the target do not share the same three-digit primary SIC code. The dependent variable in Panel A is the acquirer's cumulative abnormal return over a symmetric three-day event window around the announcement day (−1,+1). We use the market model with a 253-day estimation window and require a minimum of 100 valid returns for the acquirer to be included in the sample. We use a 63-day gap between the estimation window and the event window to ensure that the anticipation of the event does not affect the estimation of the expected return. The dependent variable in Panel B is the average of the cumulative abnormal returns across all firms within the same three-digit SIC industry as the acquirer's industry (excluding the acquirer), i.e., the rivals. The cumulative abnormal return of the rivals is calculated analogously over the same event window. Dependency of top five dependent customer industries is the average of the customer input coefficients of the top five dependent customer industries. Following Shahrur (2005), customer input coefficient is based on the Use table of the benchmark accounts from the Bureau of Economic Analysis. Customer input coefficient is defined as the dollar amount of the acquirer industry's output sold to the customer industry divided by the total output of the customer industry. For all acquisitions, target size relative to the acquirer is defined as the target deal value to the market value of the acquirer and is winsorized at the 1% and 99% levels. High industry concentration is defined in Table 2. We exclude acquisitions where the deal value is less than 2% of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A						
The dependent variable is the acquirer's CAR (−1,+1)	Horizontal acquisitions					Cross-industry acquisitions
	(1)	(2)	(3)	(4)	(5)	(6)
IPS	0.2651***	0.2709*	−0.1101	0.2527***	0.2456**	0.1469
	(0.008)	(0.070)	(0.451)	(0.009)	(0.013)	(0.169)
High industry concentration		−0.9093**				
		(0.033)				
IPS × High industry concentration		1.5745**				
		(0.037)				
Dependency of top five dependent customer industries			−0.0169**			
			(0.011)			
IPS × Dependency of top five dependent customer industries			0.0347***			
			(0.007)			
Target size relative to median book value of assets in industry				−0.0028**		
				(0.016)		
IPS × Target size relative to median book value of assets in industry				0.0052**		
				(0.018)		
Target size relative to median market value of equity in industry					−0.0069**	
					(0.033)	
IPS × Target size relative to median market value of equity in industry					0.0128**	
					(0.037)	
Acquirer log (book assets)	−0.0078*	−0.0035	−0.0088**	−0.0078*	−0.0077*	0.0050
	(0.063)	(0.289)	(0.027)	(0.057)	(0.058)	(0.397)
Target size relative to acquirer	0.0254***	0.0248**	0.0249**	0.0269***	0.0273***	0.0332**
	(0.006)	(0.023)	(0.024)	(0.004)	(0.004)	(0.016)
Log (industry sales)	−0.0045	−0.0095	−0.0068	−0.0042	−0.0045	−0.0001
	(0.445)	(0.371)	(0.245)	(0.472)	(0.432)	(0.988)
Cash payment indicator	0.0122***	0.0143***	0.0118***	0.0118***	0.0117***	0.0074
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.154)
Stock payment indicator	0.0020	0.0045	0.0003	0.0030	0.0028	−0.0139
	(0.722)	(0.435)	(0.950)	(0.577)	(0.610)	(0.296)
Private target indicator	0.0363***	0.0364***	0.0374***	0.0355***	0.0354***	0.0300***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Subsidiary target indicator	0.0365***	0.0370***	0.0378***	0.0360***	0.0359***	0.0322***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	No	No	No	No
Observations	4,261	3,284	3,868	4,261	4,261	3,372
Adjusted R-squared	0.1261	0.1621	0.1300	0.1277	0.1278	0.3814

(*continued on next page*)

(a) Original Table 3, Panel A: Acquirer CARs.

Table 3
(*continued*)

Panel B		
The dependent variable is rivals' CAR (−1,+1)	(1)	(2)
IPS	0.0292**	0.0167
	(0.024)	(0.426)
High industry concentration		−0.0598***
		(0.001)
IPS × High industry concentration		0.1080***
		(0.001)
Average rival log (book assets)	−0.0021	−0.0018
	(0.130)	(0.333)
Acquirer log (book assets)	0.0000	0.0001
	(0.904)	(0.550)
Target size relative to acquirer	−0.0004	0.0002
	(0.603)	(0.805)
Log (industry sales)	0.0001	−0.0015
	(0.914)	(0.385)
Cash payment indicator	0.0002	0.0001
	(0.757)	(0.818)
Stock payment indicator	−0.0014	−0.0005
	(0.169)	(0.615)
Private target indicator	−0.0024**	−0.0022**
	(0.032)	(0.011)
Subsidiary target indicator	−0.0029***	−0.0025**
	(0.006)	(0.011)
Year fixed effects	Yes	Yes
Firm fixed effects	No	No
Industry fixed effects	Yes	Yes
Observations	4,261	3,284
Adjusted R-squared	0.0135	0.0133

(b) Original Table 3, Panel B: Rival CARs.

6.4 Table 4: Public-target acquisitions

Read:

- Table 4 repeats the acquirer and rival announcement-return tests using only public targets.
- The acquirer IPS coefficient alone is not positive in the public-target sample, but the interaction with high concentration is positive and significant.
- The interaction with dependent customer exposure is positive and significant for acquirer returns.
- The rival return panel again shows a positive interaction of IPS with high concentration.

Economic message: The market-power interpretation survives in the public-target subsample. This matters because target premiums and combined wealth gains are observable for public-target deals.

Table 4

Acquirer and rival firm public target acquisition announcement returns and IPS.

This table examines the effect of IPS on acquirers' and rival firms' acquisition announcement returns when a firm acquires a publicly traded target in its three-digit SIC industry over the 1996–2015 sample period. The dependent variable in Panel A is the acquirer's cumulative abnormal return over a symmetric three-day event window around the announcement day (−1,+1). The dependent variable in Panel B is the average of the cumulative abnormal returns across rival firms. The cumulative abnormal return of the acquirer and rivals are calculated as in Table 3. Other variables are as previously defined. We exclude acquisitions where the deal value is less than two percent of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A

The dependent variable is the acquirer's CAR (−1,+1)	(1)	(2)	(3)
IPS	−0.1006	−0.1236	−0.4681*
	(0.424)	(0.480)	(0.065)
High industry concentration		−0.3119***	
		(0.004)	
IPS × High industry concentration		0.5525***	
		(0.009)	
Dependency of top five dependent customer industries			−0.0260**
			(0.034)
IPS × Dependency of top five dependent customer industries			0.0535**
			(0.021)
Acquirer log (book assets)	0.0002	−0.0002	0.0011
	(0.925)	(0.945)	(0.515)
Target size relative to acquirer	−0.0192	−0.0159	−0.0212
	(0.174)	(0.450)	(0.216)
Log (industry sales)	−0.0394***	−0.0354**	−0.0474***
	(0.000)	(0.012)	(0.006)
Cash payment indicator	0.0276***	0.0314***	0.0241**
	(0.000)	(0.000)	(0.013)
Stock payment indicator	−0.0103*	−0.0081	−0.0113*
	(0.096)	(0.390)	(0.093)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	No	No	No
Industry fixed effects	Yes	Yes	Yes
Observations	687	564	574
Adjusted R-squared	0.0724	0.0700	0.0596

Panel B

The dependent variable is rivals' CAR (−1,+1)	(1)	(2)
IPS	0.0776**	0.0600
	(0.047)	(0.157)
High industry concentration		−0.0877***
		(0.004)
IPS × High industry concentration		0.1280**
		(0.023)
Average rival log (book assets)	0.0001	−0.0006
	(0.966)	(0.894)
Acquirer log (book assets)	0.0015***	0.0011*
	(0.007)	(0.100)
Target size relative to acquirer	−0.0011	−0.0021
	(0.535)	(0.441)
Log (industry sales)	−0.0052	−0.0060
	(0.114)	(0.194)
Cash payment indicator	0.0007	0.0005
	(0.744)	(0.803)
Stock payment indicator	0.0000	−0.0001
	(0.997)	(0.971)
Year fixed effects	Yes	Yes
Firm fixed effects	No	No
Industry fixed effects	Yes	Yes
Observations	687	564
Adjusted R-squared	0.0168	0.0122

Table 5

Wealth effects for the target and combined merged firms in public horizontal acquisitions and IPS.

This table examines the wealth effect of IPS on the target and the combined firms in horizontal acquisitions over the 1996–2015 sample period. The sample includes horizontal acquisitions where the acquirer and the target share the same three-digit primary SIC code and where the target is a publicly traded firm. The dependent variable in Panel A is the target's offer premium calculated as log(initial offer price as reported in the SDC M&A database/target share price obtained from CRSP 63 trading days prior to the deal announcement). The dependent variable in Panel B is the combined wealth gain calculated using the target offer premium. In Models (1)-(3) the combined wealth gain is value weighted (VW) and in Models (4)-(6) it is equally weighted (EW). Other variables are as previously defined. We exclude acquisitions where the deal value is less than 2% of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A

The dependent variable is the target offer premium	(1)	(2)	(3)
IPS	−0.2409	−0.8714	−2.6773***
	(0.728)	(0.175)	(0.004)
High industry concentration		−2.7449**	
		(0.035)	
IPS × High industry concentration		5.1822**	
		(0.035)	
Dependency of top dependent customer industries			−0.0997***
			(0.006)
IPS × Dependency of top dependent customer industries			0.2053***
			(0.002)
Target log (book assets)	−0.0952***	−0.1031***	−0.1012***
	(0.000)	(0.000)	(0.000)
Acquirer log (book assets)	0.0912***	0.0995***	0.0968***
	(0.000)	(0.000)	(0.000)
Target size relative to acquirer	0.2103***	0.2094***	0.2215***
	(0.000)	(0.000)	(0.000)
Log (industry sales)	−0.0313	−0.1127	0.0039
	(0.643)	(0.136)	(0.954)
Cash payment indicator	0.0278	0.0359	0.0163
	(0.286)	(0.217)	(0.547)
Stock payment indicator	−0.0260	−0.0155	−0.0158
	(0.376)	(0.663)	(0.599)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	No	No	No
Industry fixed effects	Yes	Yes	Yes
Observations	656	549	602
Adjusted R-squared	0.1158	0.1251	0.1404

Panel B

The dependent variable is the combined wealth gain

	VW combined wealth gain			EW combined wealth gain		
	(1)	(2)	(3)	(4)	(5)	(6)
IPS	−0.4616	−0.7869**	−1.5103***	−0.1933	−0.4948	−1.5121***
	(0.132)	(0.024)	(0.004)	(0.616)	(0.172)	(0.003)
High industry concentration		−0.6404**			−1.5198**	
		(0.018)			(0.020)	
IPS × High industry concentration		1.1881**			2.8499**	
		(0.018)			(0.022)	
Dependency of top dependent customer industries			−0.0385**			−0.0508***
			(0.047)			(0.004)
IPS × Dependency of top dependent customer industries			0.0800**			0.1057***
			(0.022)			(0.001)
Target log (book assets)	−0.0174***	−0.0174*	−0.0228**	−0.0502***	−0.0546***	−0.0530***
	(0.002)	(0.078)	(0.015)	(0.000)	(0.000)	(0.000)
Acquirer log (book assets)	0.0202***	0.0218*	0.0269**	0.0478***	0.0520***	0.0507***
	(0.000)	(0.076)	(0.027)	(0.000)	(0.000)	(0.000)
Target size relative to acquirer	0.1372***	0.1533***	0.1518***	0.0986***	0.1009***	0.1043***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Log (industry sales)	−0.0369**	−0.0475**	−0.0461*	−0.0358	−0.0756*	−0.0184
	(0.038)	(0.034)	(0.072)	(0.311)	(0.053)	(0.597)
Cash payment indicator	0.0199	0.0267***	0.0136	0.0267*	0.0319**	0.0217
	(0.157)	(0.005)	(0.199)	(0.052)	(0.030)	(0.125)
Stock payment indicator	−0.0072	−0.0173	−0.0104	−0.0159	−0.0104	−0.0097
	(0.583)	(0.288)	(0.354)	(0.321)	(0.607)	(0.562)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	No	No	No	No	No
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	656	549	602	656	549	602
Adjusted R-squared	0.1642	0.1824	0.1876	0.1258	0.1424	0.1558

6.5 Table 5: Target premiums and combined wealth gains

Read:

- Panel A shows that the target offer premium is increasing in the interaction of IPS and high concentration: coefficient 5.1822.
- The interaction of IPS with dependent customer exposure is also positive for target premiums.
- Panel B shows positive interactions for value-weighted and equal-weighted combined wealth gains.
- These results imply that the acquirer-target pair jointly benefits more from horizontal acquisitions when product similarity and concentration are high.

Economic message: If horizontal acquisitions create market-power rents, target shareholders can capture some of those rents through higher premiums. Combined wealth gains imply that the market values the combined firm more, not just that wealth is transferred from acquirer to target.

6.6 Table 6: Post-acquisition industry prices

Read:

- The dependent variable is the post-acquisition change in industry prices over two years.

- The interaction of IPS and high concentration is positive and significant: 1.4153 in Model 2.
- The interaction of IPS and dependent customer exposure is positive and highly significant: 0.2927 in Model 3.
- Interactions with target size are also positive, consistent with larger targets producing bigger competitive effects.

Economic message: This table is one of the most direct pieces of product-market evidence. Stock-market reactions could reflect expectations, but post-merger price increases point to actual market-power effects.

Table 6
Changes in industry prices after horizontal acquisitions and IPS.
This table shows whether IPS impacts industry prices within the two years after the acquisition. The sample includes horizontal acquisitions where the acquirer and the target share the same three-digit primary SIC code. Industry prices are based on the producer price index (PPI), published by the Bureau of Labor Statistics adjusted for inflation (using 2012 dollars as the base). The dependent variable across all models is the change in PPI in a three-digit SIC industry within the two years after the acquisition. Other variables are as previously defined. We exclude acquisitions where the deal value is less than 2% of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

The dependent variable is the post-acquisition Δ in industry prices	(1)	(2)	(3)	(4)	(5)
IPS	0.9631	0.9229	−3.8653***	0.7488	0.8064
	(0.357)	(0.383)	(0.001)	(0.487)	(0.452)
High industry concentration	−0.0486	−0.7060**	−0.0827	−0.0526	−0.0526
	(0.427)	(0.013)	(0.351)	(0.406)	(0.406)
IPS × High industry concentration		1.4153**			
		(0.021)			
Dependency of top five dependent customer industries			−0.1475***		
			(0.001)		
IPS × Dependency of top five dependent customer industries			0.2927***		
			(0.000)		
Target size relative to median book value of assets in industry				−0.0261*	
				(0.055)	
IPS × Target size relative to median book value of assets in industry				0.0520**	
				(0.049)	
Target size relative to median market value of equity in industry					−0.0318*
					(0.095)
IPS × Target size relative to median market value of equity in industry					0.0645*
					(0.083)
Acquirer log (book assets)	0.0011	0.0010	−0.0003	0.0004	0.0002
	(0.685)	(0.698)	(0.918)	(0.901)	(0.952)
Target size relative to acquirer	0.0038	0.0044	0.0037	0.0051	0.0042
	(0.691)	(0.648)	(0.656)	(0.634)	(0.685)
Log (industry sales)	−0.0705	−0.0680	−0.0469	−0.0673	−0.0679
	(0.219)	(0.235)	(0.363)	(0.244)	(0.238)
Private target indicator	−0.0168	−0.0170	−0.0222	−0.0167	−0.0160
	(0.368)	(0.363)	(0.282)	(0.393)	(0.415)
Subsidiary target indicator	0.0105	0.0098	0.0059	0.0112	0.0120
	(0.282)	(0.317)	(0.566)	(0.277)	(0.238)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,417	1,417	1,393	1,417	1,417
Adjusted R-squared	0.3350	0.3351	0.3691	0.3359	0.3353

Table 7
Dependent customers' and suppliers' acquisition announcement returns and IPS.
This table examines the effect of IPS on the acquisition announcement returns of firms in dependent customer and supplier industries when the acquirer makes a horizontal acquisition over the 1996–2015 sample period. The sample includes horizontal acquisitions where the acquirer and the target share the same three-digit primary SIC code. The dependent variable in Panel A is the average cumulative abnormal returns across all firms in the top five dependent customer industries, and in Panel B is the average cumulative abnormal returns across all firms in the top five dependent supplier industries. The cumulative abnormal return of the firms is calculated analogously to Table 3 over the same event window. To identify the dependent customer and dependent supplier industries, we rely on the Use table of the benchmark input-output (IO) accounts published by the Bureau of Economic Analysis. Following Shahrur (2005), for each pair of customer-acquirer industries we calculate the Customer Input Coefficient (CIC) and for each pair of supplier-acquirer industries we calculate Supplier Percentage Sold (SPS). For each acquirer's industry, we identify the top five customer industries with the highest CIC as top dependent customer industries. Similarly, for each acquirer's industry, we identify the top five supplier industries with the highest SPS as top dependent supplier industries. Industry relative size is the average market value of the firms in the acquirer's industry relative to the average market value of the firms in the top five dependent customer (supplier) industries. Other variables are as previously defined. We exclude acquisitions where the deal value is less than two percent of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A

The dependent variable is dependent customers' CAR (−1,+1)	(1)	(2)	(3)	(4)	(5)
IPS	0.0268	0.0241	0.0305*	0.0312*	0.0484***
	(0.122)	(0.358)	(0.071)	(0.063)	(0.010)
High industry concentration		0.0791***			
		(0.001)			
IPS × High industry concentration		−0.1536***			
		(0.001)			
Target size relative to median book value of assets in industry			0.0060*		
			(0.058)		
IPS × Target size relative to median book value of assets in industry			−0.0124**		
			(0.043)		
Target size relative to median market value of equity in industry				0.0021**	
				(0.029)	
IPS × Target size relative to median market value of equity in industry				−0.0044**	
				(0.018)	
Industry relative size (acquirer industry to dep. customer industry)					0.0078***
					(0.003)
IPS × Industry relative size (acquirer industry to dep. customer industry)					−0.0153***
					(0.007)
Dep. customers avg. log (book assets)	−0.0012	−0.0009	−0.0012	−0.0011	−0.0009
	(0.255)	(0.351)	(0.242)	(0.271)	(0.207)
Acquirer log (book assets)	0.0001	0.0000	0.0001	0.0002	0.0001
	(0.697)	(0.932)	(0.682)	(0.614)	(0.652)
Target size relative to acquirer	0.0004	0.0007	0.0004	0.0004	0.0005
	(0.587)	(0.387)	(0.610)	(0.614)	(0.481)
Log (industry sales)	−0.0024*	−0.0031*	−0.0024*	−0.0026**	−0.0025**
	(0.055)	(0.084)	(0.055)	(0.042)	(0.043)
Cash payment indicator	0.0002	0.0003	0.0002	0.0002	0.0002
	(0.845)	(0.763)	(0.870)	(0.859)	(0.833)
Stock payment indicator	−0.0011	−0.0014	−0.0011	−0.0011	−0.0012
	(0.155)	(0.123)	(0.162)	(0.157)	(0.129)
Private target indicator	−0.0006	−0.0008	−0.0007	−0.0007	−0.0006
	(0.429)	(0.361)	(0.360)	(0.324)	(0.451)
Subsidiary target indicator	−0.0009	−0.0002	−0.0009	−0.0010	−0.0009
	(0.346)	(0.872)	(0.322)	(0.259)	(0.322)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	3,656	2,927	3,656	3,656	3,656
Adjusted R-squared	0.0421	0.0535	0.0429	0.0428	0.0449

Panel B

The dependent variable is dependent suppliers' CAR (−1,+1)	(1)	(2)	(3)	(4)	(5)
IPS	−0.0092	−0.0144	−0.0034	−0.0021	0.0069
	(0.505)	(0.501)	(0.813)	(0.882)	(0.473)
High industry concentration		0.0578***			
		(0.007)			
IPS × High industry concentration		−0.1136***			
		(0.006)			
Target size relative to median book value of assets in industry			0.0213**		
			(0.026)		
IPS × Target size relative to median book value of assets in industry			−0.0429**		
			(0.028)		
Target size relative to median market value of equity in industry				0.0210*	

(*continued on next page*)

6.7 Tables 7 and 8: Dependent customers/suppliers and acquirer performance

Read: Table 7:

- Dependent customer returns are negatively affected by $IPS \times HighConc$: coefficient -0.1536.
- Dependent supplier returns are also negatively affected by $IPS \times HighConc$: coefficient -0.1136.
- The negative effect is stronger when the target is large relative to the industry and when the acquirer industry is large relative to the dependent industry.
- The paper reports that in concentrated industries a one-standard-deviation increase in IPS lowers dependent customer and supplier announcement returns by about 19 and 32 basis points, respectively.

Read: Table 8:

- The interaction $IPS \times Acquirer$ is positive for price-cost margins.

- There is limited evidence of reduced operating expenses as the source of gains.
- Capital expenditure and advertising ratios decline in some specifications, but the main clear result is on margins.

Economic message: Customers and suppliers lose when the deal is predicted to increase market power. Acquirer performance tests suggest that the gains are not primarily explained by broad cost synergies.

Table 7
Dependent customers' and suppliers' acquisition announcement returns and IPS.

This table examines the effect of IPS on the acquisition announcement returns of firms in dependent customer and supplier industries when the acquirer makes a horizontal acquisition over the 1996–2015 sample period. The sample includes horizontal acquisitions where the acquirer and the target share the same three-digit primary SIC code. The dependent variable in Panel A is the average cumulative abnormal returns across all firms in the top five dependent customer industries, and in Panel B is the average cumulative abnormal returns across all firms in the top five dependent supplier industries. The cumulative abnormal return of the firms is calculated analogously to Table 3 over the same event window. To identify the dependent customer and dependent supplier industries, we rely on the Use table of the benchmark input-output (IO) accounts published by the Bureau of Economic Analysis. Following Shahrur (2005), for each pair of customer-acquirer industries we calculate the Customer Input Coefficient (CIC) and for each pair of supplier-acquirer industries we calculate Supplier Percentage Sold (SPS). For each acquirer's industry, we identify the top five customer industries with the highest CIC as top dependent customer industries. Similarly, for each acquirer's industry, we identify the top five supplier industries with the highest SPS as top dependent supplier industries. Industry relative size is the average market value of the firms in the acquirer's industry relative to the average market value of the firms in the top five dependent customer (supplier) industries. Other variables are as previously defined. We exclude acquisitions where the deal value is less than two percent of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A					
The dependent variable is dependent customers' CAR (−1,+1)	(1)	(2)	(3)	(4)	(5)
IPS	0.0268	0.0241	0.0305*	0.0312*	0.0484***
	(0.122)	(0.358)	(0.071)	(0.063)	(0.010)
High industry concentration		0.0791***			
		(0.001)			
IPS × High industry concentration		−0.1536***			
		(0.001)			
Target size relative to median book value of assets in industry			0.0060*		
			(0.058)		
IPS × Target size relative to median book value of assets in industry			−0.0124**		
			(0.043)		
Target size relative to median market value of equity in industry				0.0021**	
				(0.029)	
IPS × Target size relative to median market value of equity in industry				−0.0044**	
				(0.018)	
Industry relative size (acquirer industry to dep. customer industry)					0.0078***
					(0.003)
IPS × Industry relative size (acquirer industry to dep. customer industry)					−0.0153***
					(0.007)
Dep. customers avg. log (book assets)	−0.0012	−0.0009	−0.0012	−0.0011	−0.0009
	(0.255)	(0.351)	(0.242)	(0.271)	(0.207)
Acquirer log (book assets)	0.0001	0.0000	0.0001	0.0002	0.0001
	(0.697)	(0.932)	(0.682)	(0.614)	(0.652)
Target size relative to acquirer	0.0004	0.0007	0.0004	0.0004	0.0005
	(0.587)	(0.387)	(0.610)	(0.614)	(0.481)
Log (industry sales)	−0.0024*	−0.0031*	−0.0024*	−0.0026**	−0.0025**
	(0.055)	(0.084)	(0.055)	(0.042)	(0.043)
Cash payment indicator	0.0002	0.0003	0.0002	0.0002	0.0002
	(0.845)	(0.763)	(0.870)	(0.859)	(0.833)
Stock payment indicator	−0.0011	−0.0014	−0.0011	−0.0011	−0.0012
	(0.155)	(0.123)	(0.162)	(0.157)	(0.129)
Private target indicator	−0.0006	−0.0008	−0.0007	−0.0007	−0.0006
	(0.429)	(0.361)	(0.360)	(0.324)	(0.451)
Subsidiary target indicator	−0.0009	−0.0002	−0.0009	−0.0010	−0.0009
	(0.346)	(0.872)	(0.322)	(0.259)	(0.322)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	3,656	2,927	3,656	3,656	3,656
Adjusted R-squared	0.0421	0.0535	0.0429	0.0428	0.0449

Panel B					
The dependent variable is dependent suppliers' CAR (−1,+1)	(1)	(2)	(3)	(4)	(5)
IPS	−0.0092	−0.0144	−0.0034	−0.0021	0.0069
	(0.505)	(0.501)	(0.813)	(0.882)	(0.473)
High industry concentration		0.0578***			
		(0.007)			
IPS × High industry concentration		−0.1136***			
		(0.006)			
Target size relative to median book value of assets in industry			0.0213**		
			(0.026)		
IPS × Target size relative to median book value of assets in industry			−0.0429**		
			(0.028)		
Target size relative to median market value of equity in industry				0.0210*	

(*continued on next page*)

Original Table 7, first page: Dependent customer and supplier returns.

Table 7 (*continued*)

				(0.054)	
IPS × Target size relative to median market value of equity in industry				−0.0416*	
				(0.055)	
Industry relative size (acquirer industry to dep. supplier industry)					0.0056**
					(0.033)
IPS × Industry relative size (acquirer industry to dep. supplier industry)					−0.0113**
					(0.034)
Dependent supplier avg. log (book assets)	0.0008	0.0012	0.0007	0.0008	0.0005
	(0.160)	(0.103)	(0.174)	(0.150)	(0.884)
Acquirer log (book assets)	−0.0005***	−0.0005***	−0.0005*	−0.0005**	−0.0003*
	(0.006)	(0.008)	(0.053)	(0.024)	(0.065)
Target size relative to acquirer	0.0006	−0.0004	0.0006	0.0006	0.0004
	(0.469)	(0.638)	(0.533)	(0.517)	(0.586)
Log (industry sales)	−0.0002	−0.0002	−0.0002	−0.0002	0.0012
	(0.826)	(0.888)	(0.843)	(0.814)	(0.296)
Cash payment indicator	0.0007	0.0010	0.0007	0.0007	−0.0001
	(0.228)	(0.104)	(0.259)	(0.249)	(0.858)
Stock payment indicator	−0.0013*	−0.0014	−0.0014*	−0.0014*	−0.0020*
	(0.095)	(0.128)	(0.063)	(0.075)	(0.096)
Private target indicator	−0.0020**	−0.0027***	−0.0021***	−0.0021***	−0.0011
	(0.013)	(0.007)	(0.003)	(0.007)	(0.107)
Subsidiary target indicator	−0.0016**	−0.0025**	−0.0018**	−0.0017*	−0.0009
	(0.046)	(0.011)	(0.043)	(0.050)	(0.154)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	3,896	3,103	3,896	3,896	3,896
Adjusted R-squared	0.0465	0.0507	0.0478	0.0475	0.0491

(a) Original Table 7, continuation.

Table 8

The impact of IPS on acquirer firm performance.

This table examines the effect of IPS on changes in the price-cost margin and various expense ratios over the two years following an acquisition for acquirers compared to matched control firms. The acquirer indicator equals one if a firm makes a horizontal acquisition and equals zero for matched control firms, i.e., firms that do not make a horizontal acquisition. One control firm is matched to each acquiring firm. Each matched control firm operates in the same three-digit SIC industry as the acquiring firm during the year of the horizontal acquisition, and is the firm closest in size to the acquiring firm that does not make a horizontal acquisition that year. The dependent variables in Models (1)-(6) are, respectively, changes in the "price-cost margin," "cost of goods sold/sales," "selling, general, and administrative expenses/sales," "capital expenditures/assets," "research and development expenses/sales," and "advertisement expenses/sales" over the two years after a horizontal acquisition. The price-cost margin is defined as sales minus cost of goods sold scaled by sales. We exclude acquisitions where the deal value is less than 2% of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

The dependent variable is	Δ Price-cost margin	Δ COGS/Sales	Δ SGA/Sales	Δ CAPX/Assets	Δ R&D/Sales	Δ Adv./Sales
	(1)	(2)	(3)	(4)	(5)	(6)
IPS × Acquirer indicator	2.1675**	0.8123	0.2017	−0.0559**	1.0383	−0.0164*
	(0.027)	(0.707)	(0.496)	(0.020)	(0.476)	(0.089)
IPS	−1.4301	−1.9514	−0.3188	0.0035	−1.7282	−0.0045
	(0.251)	(0.342)	(0.496)	(0.947)	(0.259)	(0.641)
Acquirer indicator	−0.9320*	−0.4130	−0.0918	0.0266**	−0.4693	0.0091*
	(0.054)	(0.672)	(0.523)	(0.020)	(0.483)	(0.074)
Log (book assets)	−0.0168	0.0317**	0.0120**	−0.0001	0.0240*	0.0003
	(0.178)	(0.045)	(0.027)	(0.853)	(0.051)	(0.179)
Log (industry sales)	−0.0172	0.1869*	0.0150	0.0013	0.1225	0.0002
	(0.800)	(0.094)	(0.435)	(0.538)	(0.123)	(0.627)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,851	7,851	7,851	7,851	7,851	7,851
Adjusted R-squared	0.0063	0.0132	0.0265	0.0556	0.0110	0.0267

announcement), so these tests will reveal the incremental benefit to the merging parties. In the models in Table 8 the

ation, and there is not a clear prediction on the interactive effect of industry product similarity and concentration on

(b) Original Table 8: Acquirer performance.

6.8 Table 9: Antitrust challenges

Read:

- Panel A shows that challenge rates are much higher for high-IPS deals: 13.4% for above-median IPS versus 2.9% for low IPS; 16.6% for top-quartile IPS versus 5.2% for low IPS.
- Panel B shows a positive IPS coefficient in the challenge regression: 2.0168 in Model 1.
- The interaction of IPS and high concentration is positive: 4.9614 in Model 2.
- Interactions with customer dependency, supplier dependency, and target relative size are positive.
- Panel C shows that IPS has stronger effects on challenge probability when adjusted acquirer returns, adjusted rival returns, or adjusted combined wealth gains are high.

Economic message: Regulators appear to identify the same economic pattern as investors: high-IPS horizontal deals in concentrated industries are more likely to create anticompetitive effects.

Table 9
Challenged acquisitions and IPS.
This table examines whether IPS affects the likelihood of an acquisition being challenged by the Federal Trade Commission (FTC) or the Department of Justice (DOJ) over the 1996–2015 sample period. We hand-collect data on all challenged deals from the FTC and DOJ websites. The sample includes horizontal acquisitions where the acquirer and the target share the same three-digit primary SIC code and where the target is a publicly traded firm. We exclude acquisitions where the deal value is less than 2% of the market value of the acquirer. Also, we require non-missing values for the control variables used in the regression models. This yields a sample of 59 challenged acquisitions with publicly traded targets in which 54 acquisitions are completed and 5 are withdrawn. Panel A reports the percent of horizontal acquisitions that are challenged in low versus high IPS industries and reports significance levels for whether they are significantly different across the low and high IPS subsamples. Panel B reports multivariate evidence on the effect of IPS on the likelihood an acquisition is challenged. Dependency of top five dependent supplier industries is the degree to which the sales of the top five dependent supplier industries rely on the acquirer's industry. This measure is the average of Supplier Percentage Sold across the top five dependent supplier industries. Panel C examines the impact of the acquirer's and rival firms' announcement returns and the wealth gain of the combined merged firms on the effect of IPS on the likelihood an acquisition is subsequently challenged. Models (1)-(4) in Panel C interact IPS, respectively, with the adjusted announcement returns of the acquirer and rivals, and the wealth gains of the merged firm. Because the acquirer's, rival firms', and the target firm's announcement returns and the premium paid for the target are endogenous to the likelihood the acquisition is subsequently challenged, we adjust announcement returns and the wealth gain of the merged firms by conducting a two-stage regression analysis. In the first stage, we instrument the announcement returns or the wealth gain to the merged firm with exogenous variables that are directly unrelated to the likelihood of the acquisition being challenged. The instruments are (a) a surprise deal dummy, which equals one if there is no horizontal acquisition of a publicly traded target in the industry with the horizontal acquisition during the three years before the deal announcement, and zero otherwise; (b) a tender offer dummy; (c) a hostile takeover dummy; (d) industry-adjusted stock payment percentage, which equals the deal's consideration paid in stock minus the average consideration paid in stock for all horizontal acquisitions in the industry with the horizontal acquisition during the year of the deal announcement; and (e) the acquirer's prior-year operating income before depreciation/book assets. In the second-stage models, we use the residuals from the first-stage model as the measures for adjusted announcement returns and the wealth gain to the merged firm and interact the residuals with IPS. The acquirer's and rivals' announcement returns in Models (1) and (2) are calculated as in Table 3, and the combined wealth gain in Models (3) and (4) are calculated as in Table 5. Other variables are as previously defined. We exclude acquisitions where the deal value is less than 2% of the market value of the acquirer. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A

	Observations	% of challenged acquisitions
Low IPS	377	2.9%
High IPS (above median)	358	13.4%
Diff = (% in High IPS) - (% in Low IPS)		10.5%
		(0.000)
Low IPS	554	5.2%
High IPS (top quartile)	181	16.6%
Diff = (% in High IPS) - (% in Low IPS)		11.4%
		(0.000)

Panel B

The dependent variable is an indicator for challenged acquisition	(1)	(2)	(3)	(4)	(5)	(6)
IPS	2.0168***	1.9142***	0.5016	1.0747	1.2764	1.1679
	(0.006)	(0.007)	(0.495)	(0.142)	(0.119)	(0.161)
High industry concentration	0.1599	−2.5495*	−0.0235	0.3033	0.1428	0.1646*
	(0.102)	(0.067)	(0.719)	(0.205)	(0.162)	(0.093)
IPS × High industry concentration		4.9614**				
		(0.048)				
Dependency of top five dependent customer industries			−0.1050***			
			(0.004)			
IPS × Dependency of top five dependent customer industries			0.2003***			
			(0.002)			
Dependency of top five dependent supplier industries				−0.0602**		
				(0.025)		
IPS × Dependency of top five dependent supplier industries				0.1247**		
				(0.021)		
Target size relative to median book value of assets in industry					−0.2640**	
					(0.025)	
IPS × Target size relative to median book value of assets in industry					0.5872**	
					(0.012)	
Target size relative to median market value of equity in industry						−0.3578**
						(0.015)
IPS × Target size relative to median market value of equity in industry						0.7931***
						(0.007)
Acquirer log (book assets)	0.0372***	0.0372***	0.0303**	0.0322***	0.0223*	0.0245**
	(0.004)	(0.004)	(0.011)	(0.007)	(0.057)	(0.041)
Target size relative to acquirer	0.0466*	0.0475*	0.0346	0.0370	0.0201	0.0152
	(0.085)	(0.079)	(0.181)	(0.129)	(0.333)	(0.492)
Log (industry sales)	0.0645	0.0655	0.0597	0.0351	0.0498	0.0513
	(0.153)	(0.143)	(0.149)	(0.379)	(0.228)	(0.212)
Market share	0.0927	0.1109	0.5400**	0.3415	0.0824	0.0927
	(0.727)	(0.674)	(0.043)	(0.256)	(0.760)	(0.732)
Cash payment indicator	−0.0212	−0.0209	−0.0226	−0.0112	−0.0022	−0.0074
	(0.476)	(0.474)	(0.410)	(0.653)	(0.948)	(0.817)
Stock payment indicator	0.0285	0.0282	0.0385	0.0378	0.0175	0.0233
	(0.339)	(0.335)	(0.203)	(0.205)	(0.485)	(0.373)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	735	735	646	646	735	735
Adjusted R-squared	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

(a) Original Table 9, Panels A and B: Challenge probabilities.

Table 9
(*continued on next page*)

Panel C

The dependent variable is an indicator for challenged acquisition	(1)	(2)	(3)	(4)
IPS	1.5864**	1.7747**	1.6163	1.5382*
	(0.031)	(0.038)	(0.103)	(0.083)
High industry concentration	0.0846	0.1164	0.0407	0.0482
	(0.567)	(0.464)	(0.750)	(0.691)
Adjusted acquirer's CAR	−4.7456***			
	(0.008)			
IPS × Adjusted acquirer's CAR	9.9519***			
	(0.007)			
Adjusted rivals' CAR		−11.5384***		
		(0.008)		
IPS × Adjusted rivals' CAR		24.2953***		
		(0.005)		
Adjusted VW combined wealth gain (target offer premium)			−3.8091**	
			(0.046)	
IPS × Adjusted VW combined wealth gain (target offer premium)			8.1912**	
			(0.041)	
Adjusted VW combined CAR (−1,+1)				−4.8387**
				(0.022)
IPS × Adjusted VW combined CAR (−1,+1)				10.2020**
				(0.021)
Market share	0.2819	0.1978	0.1172	0.1425
	(0.284)	(0.467)	(0.746)	(0.691)
Target log (book assets)	0.0382***	0.0406***	0.0413***	0.0430***
	(0.006)	(0.008)	(0.008)	(0.007)
Acquirer log (book assets)	0.0514	0.0409	0.0178	0.0304
	(0.103)	(0.184)	(0.611)	(0.337)
Log (industry sales)	0.0627	0.0604	0.0794	0.0766
	(0.244)	(0.274)	(0.192)	(0.175)
Cash payment indicator	−0.0608	−0.0575	−0.0740*	−0.0709
	(0.124)	(0.134)	(0.080)	(0.101)
Stock payment indicator	−0.0061	−0.0026	−0.0172	−0.0145
	(0.842)	(0.931)	(0.586)	(0.633)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Observations	610	610	504	513
Adjusted R-squared	0.1620	0.1512	0.1555	0.1568

(b) Original Table 9, Panel C: Challenge probability and market-power proxies.

6.9 Table 10: IP-focused and commodity-based industries

Read:

- Panel A shows that the triple interaction of IPS, high concentration, and IP-focused industry is negative for acquirer returns, rival returns, and price changes.
- This means the main effects are weaker in IP-focused industries.
- Panel B shows that the triple interaction with commodity-based industries is positive for acquirer and rival returns.
- This means the main effects are stronger in commodity-based industries.

Economic message: The heterogeneity results sharpen the mechanism. Product similarity matters most where products are naturally comparable and coordination is easier, and it matters less where innovation and differentiation are central.

Table 10
Intellectual property-focused and commodity-based industries and IPS.

This table examines whether the interactive effect of IPS and high industry concentration is different for firms in intellectual property-(IP) focused industries or commodity-based industries compared to that in other industries. In Panel A, to consider the degree to which an industry is IP-focused, we use data on patents issued to US firms by the United States Patent and Trademark Office (USPTO) and link these data to Compustat data. Following Acharya et al. (2014), to calculate a firm-level measure for the extent to which a firm is IP-focused, we divide the number of new patents issued to a firm in a given year by the number of employees of the firm that year. To estimate the degree to which over a year a given three-digit SIC industry is IP-focused, we calculate the mean value of the number of new patents issued to a firm by the number of its employees across all the firms in the industry. In Panel B we use an indicator for commodity-based industries. This variable is an indicator equal to one if a firm is in a metal mining industry (two-digit SIC code of 10), a coal or oil and gas industry (two-digit SIC of 12 or 13), or the manufacturing sector (two-digit SIC from 20 to 39) and equal to zero otherwise. The indicator for commodity-based industries also gets a value of zero if a firm's three-digit SIC industry is IP-focused, as measured by whether the industry mean for number of new patents issued scaled by number of employees is in the top quintile across all industries that year. The control variables in Models (1) and (2) of both panels, respectively, are the same as those in Model (2) of panels A and B of Table 3. The control variables in Panel A, Model (3) are the same as those in Model (2) of Table 6. Industry is defined at the three-digit SIC level. The standard errors are corrected for heteroskedasticity and clustering at the industry level. P-values are reported in parentheses and ***, **, and * indicate whether the coefficient estimate is significantly different from zero at the 1%, 5%, or 10% level.

Panel A

The dependent variable is	Acquiror's CAR (−1,+1) (1)	Rivals' CAR (−1,+1) (2)	Post-acquisition Δ in industry prices (3)
IPS × High industry concentration × IP-focused industry	−0.3143***	−0.0180**	−0.8042**
	(0.002)	(0.032)	(0.039)
IPS × High industry concentration	1.6944**	0.0900**	2.9855***
	(0.028)	(0.038)	(0.000)
IPS × IP-focused industry	0.0125	0.0012	0.1105***
	(0.152)	(0.258)	(0.010)
High industry concentration × IP-focused industry	0.1883***	0.0063	0.4234**
	(0.000)	(0.140)	(0.042)
IPS	0.2038	0.0120	−0.6601
	(0.206)	(0.615)	(0.445)
High industry concentration	−0.9814**	−0.0477**	−1.4929***
	(0.024)	(0.045)	(0.000)
IP-focused industry	−0.0052	−0.0006	−0.0545**
	(0.292)	(0.230)	(0.021)
Controls	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	No	No
Industry fixed effects	No	Yes	Yes
Observations	3,284	3,284	1,417
Adjusted R-squared	0.1120	0.0116	0.3463

Panel B

The dependent variable is	Acquiror's CAR (−1,+1) (1)	Rivals' CAR (−1,+1) (2)
IPS × High industry concentration × Commodity-based industry	0.5209**	0.1076**
	(0.039)	(0.027)
IPS × High industry concentration	1.4098*	0.1006**
	(0.090)	(0.013)
IPS × Commodity-based industry	−0.5638	−0.0731
	(0.103)	(0.205)
High industry concentration × Commodity-based industry	−0.2622**	−0.0454*
	(0.048)	(0.088)
IPS	0.2933*	0.0203
	(0.059)	(0.342)
High industry concentration	−0.8178*	−0.0572**
	(0.081)	(0.014)
Commodity-based industry	0.2476	0.0385
	(0.146)	(0.161)
Controls	Yes	Yes
Year fixed effects	Yes	Yes
Firm fixed effects	Yes	No
Industry fixed effects	No	Yes
Observations	3,284	3,284
Adjusted R-squared	0.1106	0.0119

Original Table 10: IP-focused and commodity-based industries and IPS.

7 Short summary

- The paper studies whether horizontal acquisitions reduce industry competition.
- The key argument is that product similarity is necessary for concentration to be informative.
- IPS measures within-industry product similarity using 10-K textual analysis.
- IPS is validated through correlations with homogeneity, product-market fluidity, firm-pair similarity, and margins.
- High IPS raises the probability of a horizontal acquisition, especially in concentrated industries.
- High IPS does not predict cross-industry acquisitions.
- Acquirer announcement returns are higher for high-IPS horizontal acquisitions.
- Rival returns are higher, especially in high-IPS/high-concentration industries.
- Target premiums and combined wealth gains are higher in the same settings.
- Industry prices rise after horizontal acquisitions when IPS and concentration are both high.
- Dependent customers and suppliers experience negative announcement returns.
- Acquirer performance evidence points more to margin increases than cost synergies.
- FTC/DOJ challenges are more likely for high-IPS horizontal acquisitions.
- Effects are weaker in IP-focused industries and stronger in commodity-based industries.
- Concentration alone does not explain the outcomes, which helps reconcile mixed prior evidence.

8 Conclusion

8.1 Core conclusion

Fathollahi, Harford, and Klasa show that horizontal acquisitions are more likely to have anticompetitive effects when the relevant industry is both concentrated and product-similar. Product similarity is the missing conditioning variable in much of the prior horizontal-merger literature. Concentration alone is insufficient because it does not measure whether firms sell close substitutes.

8.2 Economic intuition

The intuition is that product similarity changes the competitive meaning of concentration. In homogeneous-product industries, firms can more easily coordinate on prices or output because products are directly comparable and deviations are easier to detect. A horizontal acquisition removes a competitor and can make coordination more profitable. This benefits acquirers, targets, and rivals but hurts customers and suppliers. In differentiated-product industries, concentration is less informative because competition is local and product-specific.

8.3 Takeaway

The paper provides a clear lesson for empirical corporate finance and antitrust analysis: do not study horizontal acquisitions using concentration alone. Product similarity determines when concentration is likely to translate into market power. The results make IPS a valuable tool for studying M&A, product market competition, antitrust enforcement, and the corporate policies of firms operating in similar-product industries.